

Predicting the Interest Rate Volatility Term Structure via PCA Decomposition and Regularised Models

A Study of the U.S. Treasury Realized Volatility, from 2006 to 2026

1 Introduction

The realized volatility term structure of interest rates describes the realized volatility across tenors. It is a central input in the pricing and risk measure of interest rate derivatives, where the vega profile of a book is directly exposed to shifts in this structure across tenors. Anticipating whether short-tenor volatility will outpace long-tenor volatility, or whether a broad expansion will lift the entire curve allow for better risk management.

This work is based on two pieces of academic literature. [Litterman & Scheinkman \(1991\)](#) establish that yield curve dynamics decompose into three orthogonal factors: level, slope, and curvature. Each has distinct driver set. We extend this decomposition to the realized volatility curve. [Corsi \(2009\)](#) demonstrates that autoregressive features (daily, weekly, and monthly averages) substantially improve realized volatility forecasts by capturing the different time horizons at which market participants react. Combining both, we decompose the volatility curve via PCA and use a large set of features that includes macro indicators, yield curve dynamics, and persistence lags on the PCA scores themselves.

2 Data and Target Construction

2.1 Realized Volatility Proxies

Daily Treasury yields for the 2Y, 5Y, 10Y, and 30Y tenors are sourced from FRED (series DGS2, DGS5, DGS10, DGS30) over January 2005 to April 2026. For each tenor, realized volatility is defined as the twenty-day rolling standard deviation of daily yield changes. These series serve both as a measure of realized volatility but also as a proxies for ATM short-term implied volatility, under the assumption that realized and implied volatility co-move over the estimation window.

2.2 PCA Decomposition

PCA is applied to the standardized matrix of four realized volatility series. Three components are retained, explaining 99.7% of total variance: PC1 (82.0%), PC2 (15.7%), and PC3 (1.9%). As shown in Figure 1, the decomposition recovers the factor structure of [Litterman & Scheinkman \(1991\)](#) applied to the volatility curve: PC1 carries uniformly positive loadings across all tenors (parallel volatility shift, i.e. level); PC2 has monotonically decreasing loadings from short to long tenors (slope, i.e. short-tenor volatility vs. long-tenor volatility); PC3 exhibits a humped

profile peaking at the 5Y and 10Y tenors (curvature). PC1 correlates 0.871 with the MOVE Index in-sample, confirming that it captures the curve parallel shift effect.

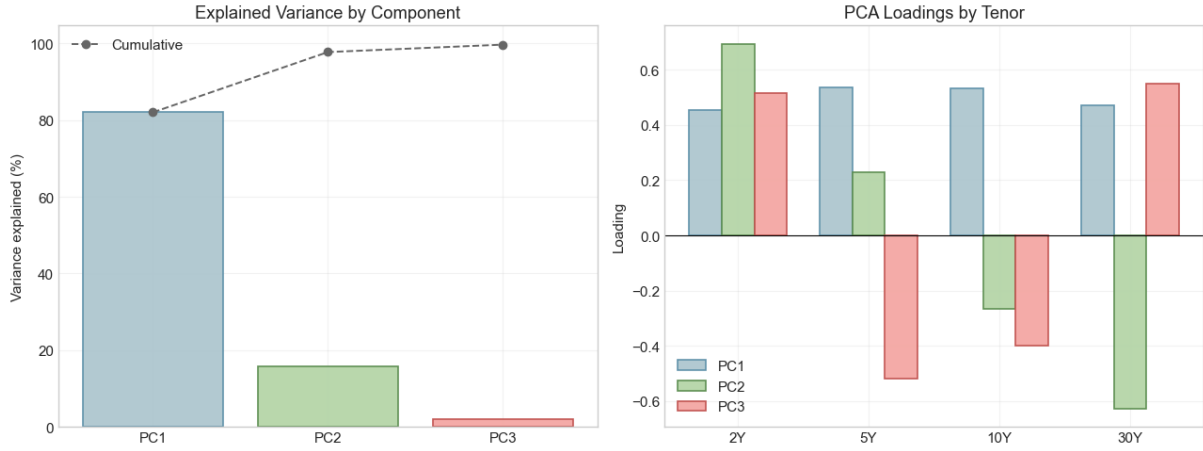


Figure 1: PCA loadings on the four-tenor realized volatility curve.

2.3 Target Construction

Instead of the realized volatility targets at each tenor, the targets are PCA score levels 20 days in the future, at $t + 20$:

$$y_k(t) = \text{PC}_k(t + 20), \quad k \in \{1, 2, 3\}$$

The twenty-day horizon matches the volatility estimation window, ensuring no overlap between successive target periods. This is to avoid autocorrelation bias in model evaluation. The three targets are orthogonal by construction (PCA produces uncorrelated components), which justifies training three independent models rather than a joint multivariate one.

3 Feature Engineering

The feature set contains 41 variables, built from the downloaded series, and organised into seven groups. Each group has a different economic interpretations:

- *Yield curve structure* — 10Y rate level, 5-day momentum, 10-2Y slope and 30-5Y slope
- *Realized volatility* — 20-day rolling volatility for 2Y and 10Y tenors, and their ratio as a measure of curve change in realized volatility
- *Market sentiment* — VIX, SKEW, MOVE Index
- *Macroeconomic indicators* — Fed Funds rate, 5Y inflation breakeven, credit spread, and the Fed lag (FED funds - 2Y rate slope) as a measure of macro policy relative to the front end
- *Curve dynamics* — 5-day and 20-day changes in the 2–10Y slope, and relative tenor moves
- *Mean-reversion z-scores* — 6-month rolling z-scores for VIX, MOVE, slope, and each PC score, capturing the distance of each variable from its recent mean
- *Autoregressive lags* — for each PC score: yesterday's value ($t - 1$), 5-day average, and 22-day average, capturing the daily, weekly, and monthly persistence horizons.

4 Models and Results

Data are split chronologically between training (2006–2021) and testing (2022–2026) sets. The test period coincides with the Fed tightening cycle and the 2022–2023 rate volatility spike, making it a conservative and realistic test environment. Three metrics are reported throughout: RMSE relative to a naive persistence baseline (volatility does not change); the Information Coefficient (IC), defined as the Spearman rank correlation between predictions and realizations; and directional accuracy with ROC-AUC. The IC is the primary metric for practical relevance, as a model that correctly ranks high-volatility days above low-volatility days is actionable. Robustness is assessed via ten-fold expanding walk-forward validation.

4.1 Elastic Net

Elastic Net is a linear regression model that minimizes a penalized OLS objective that combines L1 and L2 penalties. The mixing parameter $\rho \in [0, 1]$ is selected by five-fold time-series cross-validation within the training set. The selected ρ is itself a diagnostic result: $\rho = 1$ indicates that the relevant features are sufficiently orthogonal for L1 alone to produce stable estimates; $\rho < 1$ signals that correlated predictors benefit from shrinkage.

On PC1 and PC3, cross-validation selects $\rho = 1.0$ (pure L1). On PC2, $\rho = 0.5$ is selected: yield curve slope variables, monetary policy, and lagged values are sufficiently correlated that joint L2 shrinkage improves out-of-sample performance. This result is economically interpretable: the drivers of volatility slope, namely FED policy, current shape, and momentum are highly collinear, and regularisation correctly prevents the model from concentrating all weight.

4.2 Gradient Boosting

Gradient Boosting serves as a non-linear benchmark, motivated by the possibility that interactions between macro variables might add predictive value beyond what a linear model can capture. To prevent overfitting, a two-pass procedure is used. In the first pass, a quick regularised GB model is fitted on all 41 features and feature importances are computed via Mean Decrease in Impurity (MDI). Features with above-average MDI importance are selected. In the second pass, a tuned model is trained on the selected subset via grid-search cross-validation over the hyperparameters.

Cross-validation selects moderately regularised configurations across all three components: max depth of 3–4, learning rate of 0.01–0.05 (requiring many trees to converge), and minimum sample leaf of 5–20. These settings reflect the low signal-to-noise environment: the CV penalty for complexity is high enough that the optimal model remains conservative. The model evaluation shows that train-test RMSE gaps are $1.5\times$ on PC1, $1.8\times$ on PC2, and $1.3\times$ on PC3, which are within an acceptable range of overfitting.

4.3 Model Comparison

The model evaluation results are reported in Table 1. Elastic Net is selected on all three components based on IC. The most striking result is PC2: Gradient Boosting achieves an IC of -0.207 , meaning it systematically inverts the correct ranking of slope moves, regardless of the regularisation level applied. This reflects a fundamental signal-to-noise issue: PC2 explains only 16% of volatility curve variance, and the non-linear model overfits to training noise in the

training set rather than extracting patterns. The linear model, constrained by L1 penalty, is more robust to this regime.

Table 1: Model Performance by PCA Component vs. Persistence Baseline (test set)

Component	Elastic Net			Gradient Boosting			Persist.	Selected
	RMSE	IC	Skill	RMSE	IC	Skill	RMSE	
PC1 (level, 82%)	1.399	0.649	0.187	1.437	0.598	0.164	1.720	ENet
PC2 (slope, 16%)	0.662	0.231	0.160	0.826	-0.207	-0.048	0.788	ENet
PC3 (curv., 2%)	0.247	0.122	0.217	0.249	0.064	0.212	0.315	ENet

On PC1, both models beat the persistence baseline (positive skill), confirming that the feature set carries information beyond the current volatility level. The IC of 0.649 on PC1, the dominant factor explaining 82% of variance, is the main result of this work. In fact, it represents a strong ranking signal on the most economically significant component.

Table 2: Elastic Net Directional Prediction Accuracy and Walk-Forward RMSE

Component	Accuracy	ROC-AUC	WF RMSE (ENet)	WF RMSE (GB)
PC1 (level)	72.5%	0.784	1.852	1.849
PC2 (slope)	70.9%	0.782	0.663	0.797
PC3 (curv.)	68.7%	0.777	0.273	0.337

In addition, Table 2 shows the directional results of both models. The three components exceed 68% directional accuracy with ROC-AUC above 0.77. The PC2 result is particularly instructive: despite modest RMSE-based metrics ($R^2 = 0.065$), the model correctly calls the direction of slope moves 71% of the time. This dissociation between RMSE and directional accuracy reflects attenuation bias: the model predicts the conditional mean, which is smoother than the realized series, but still correctly ranks up versus down volatility move days. This is further illustrated by the ROC plotted in Figure 2. Walk-forward results confirm the test-set ranking: Elastic Net is dominant on PC2 and PC3, where the walk-forward RMSE gap over Gradient Boosting is 20% and 23% respectively.

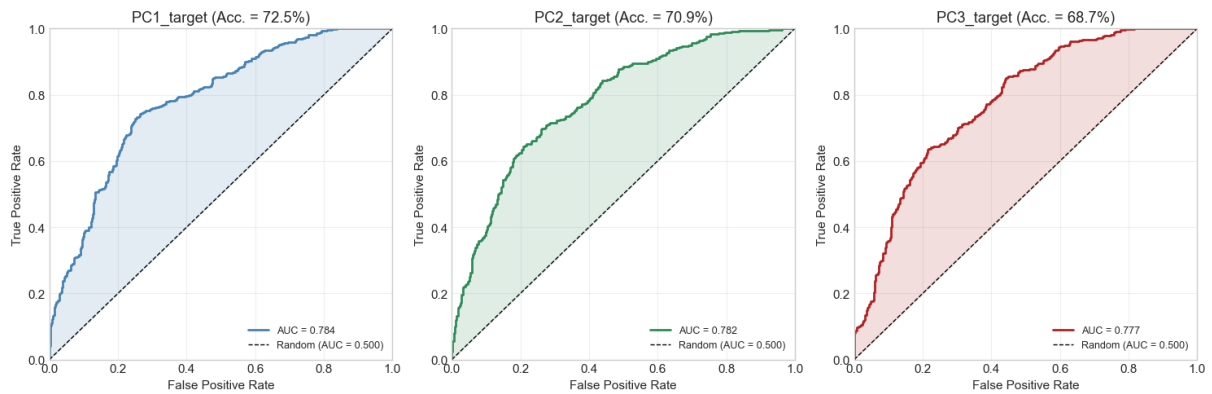


Figure 2: Elastic Net ROC curves (test set). The shaded area is the gain over a random classifier.

5 Elastic Net: Further Analysis

5.1 Retained Coefficients and Economic Interpretation

The L1 penalty drives most coefficients to exactly zero, making the retained features directly interpretable as the model’s effective information set. Figure 3 below shows the value of each non-negative coefficients.

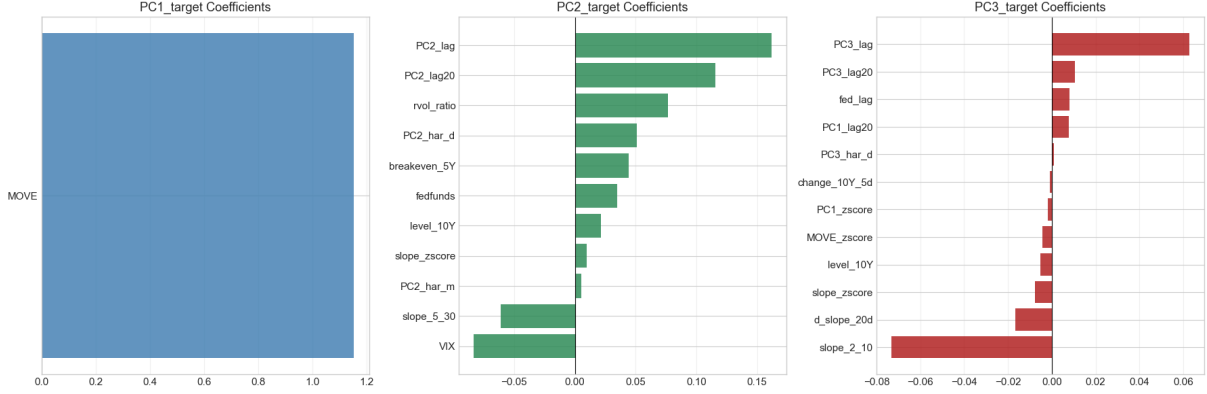


Figure 3: Non-zero Elastic Net coefficients by PCA component.

PC1 (level) — **IC = 0.649, accuracy = 72.5%, skill = 0.187.** Only 1 out of 41 features survives the L1 penalty: the MOVE Index, with a large positive coefficient (≈ 1.1). This result is surprising but economically coherent. The MOVE Index is itself a measure of aggregate implied rate volatility. Its near-exclusive selection reflects the strong regime persistence of volatility level: the dominant predictor of realized volatility over the next twenty days is the current implied volatility environment, with no additional marginal information from yield curve shape, macro variables, or autocorrelation lags once MOVE is included. This also explains the ρ parameter of 1.0: there is no collinearity to regularise away.

PC2 (slope) — **IC = 0.231, accuracy = 70.9%, skill = 0.160.** Eleven features are retained, with a more complex structure than PC1. The two largest positive coefficients belong to the current PC2 value (≈ 0.16) and the PC2 20-lags value (≈ 0.11). The persistence structure dominates, confirming that volatility slope is a persistent regime variable: yesterday and today’s shape are the top predictors of tomorrow’s. Beyond persistence, the model retains few positive coefficients, such as the short-to-long realized ratio and the 5Y breakeven inflation. On the negative side, retained features include the VIX level and the 5–30Y slope. The VIX result is interpretable: a rise in equity volatility tends to compress short-tenor rate volatility relative to long-tenor volatility, as the market focuses on near-term risk-off dynamics, the volatility curve flattens, reducing PC2.

PC3 (curvature) — **IC = 0.122, accuracy = 68.7%, skill = 0.217.** Twelve features are retained, dominated by autoregressive structure and yield curve shape. The current PC3 value carries the largest positive coefficient (≈ 0.06), reinforcing the mean-reverting nature of curvature: the current curvature level is the primary predictor of curvature twenty days ahead, with a positive sign indicating persistence rather than reversion on this horizon. On the negative side, the 2–10Y slope (≈ -0.08) is the top predictor: a steepening of the curve is associated with a reduction in volatility curvature, likely because steepening regimes concentrate uncertainty at the short end rather than in the middle. The slight positive contribution of Fed Lag (≈ 0.01)

captures the role of Fed policy stance: when the policy rate significantly exceeds the 2Y yield, near-term rate uncertainty is high, which tends to lift the middle of the volatility curve.

5.2 Regime Sensitivity

Splitting the test set into stress ($\text{MOVE} > 100$, $n = 610$ days) and calm ($\text{MOVE} < 80$, $n = 146$ days) periods reveals a sharp performance differential. Table 3 below performs this decomposition.

Table 3: Elastic Net RMSE by Market Regime (test set).

Component	Stress RMSE	Calm RMSE	Ratio
PC1 (level)	1.518	0.878	$1.73\times$
PC2 (slope)	0.969	0.190	$5.10\times$
PC2 (curvature)	0.284	0.181	$1.57\times$

The model is $1.7\times/1.5\times$ less accurate on PC1/PC2 in stress regimes, and $5.1\times$ less accurate on PC2. The PC2 result is particularly striking: volatility slope dynamics during stress periods are largely disconnected from the macro features in the training set, as the Fed’s forward guidance compresses short-tenor volatility in ways that are difficult to predict from historical patterns alone. Residual analysis confirms fat right tails in PC1 and PC2 errors: the model systematically underestimates extreme volatility expansions, a structural consequence of predicting the conditional mean. Note that the test set is dominated by stress observations (610 vs. 146 calm days), making the overall test RMSE a conservative estimate of full-cycle performance.

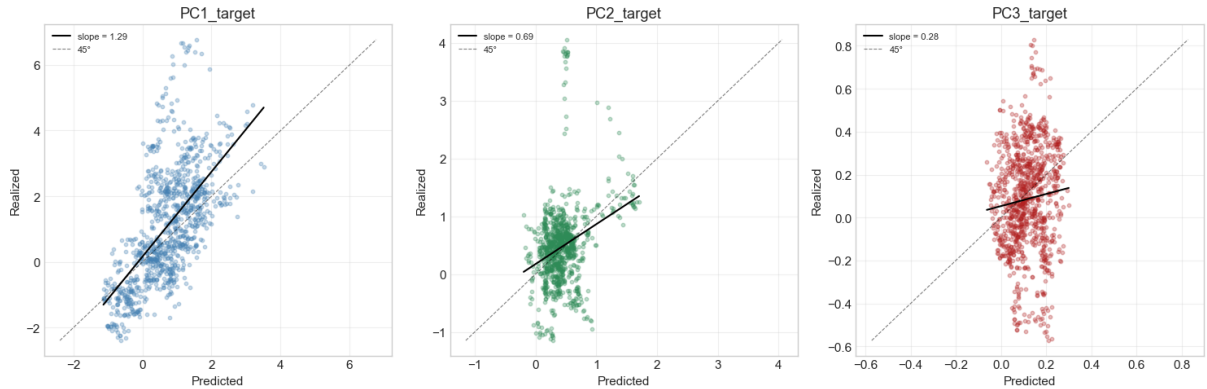


Figure 4: Elastic Net predicted vs. realized PCA score scatter plot

The scatter plot in Figure 4 further illustrates this regime sensitivity. On PC1, the regression slope of 1.29 above the 45-degree diagonal reflects the attenuation bias: the model correctly ranks observations but compresses the predicted range relative to realized values, systematically underestimating the largest vol expansions. On PC3, the near-flat regression slope (0.28) confirms that the model has limited amplitude prediction — it captures the direction but not the magnitude of curvature moves.

5.3 Volatility Curve Reconstruction

The three predicted PCA scores can be mapped back to the original tenor space via the inverse PCA transform:

$$\hat{\sigma}_{t+20} = \boldsymbol{\mu} + \mathbf{W} \hat{\mathbf{p}}_{t+20}$$

where $\hat{\mathbf{p}}_{t+20} = (\hat{PC}_1, \hat{PC}_2, \hat{PC}_3)^\top$ are the predicted scores, $\mathbf{W}$ is the 4×3 matrix of PCA loadings, and $\boldsymbol{\mu}$ is the mean realized volatility vector across the four tenors. This yields a direct prediction of the full volatility term structure (realized volatility at 2Y, 5Y, 10Y, and 30Y) at a twenty-day horizon. Figure 5 shows the predicted versus realized volatility by tenor. Because PC1 dominates (82% of variance), the shape of the reconstructed curve is primarily driven by the level prediction, with PC2 and PC3 introducing first and second-order adjustments.

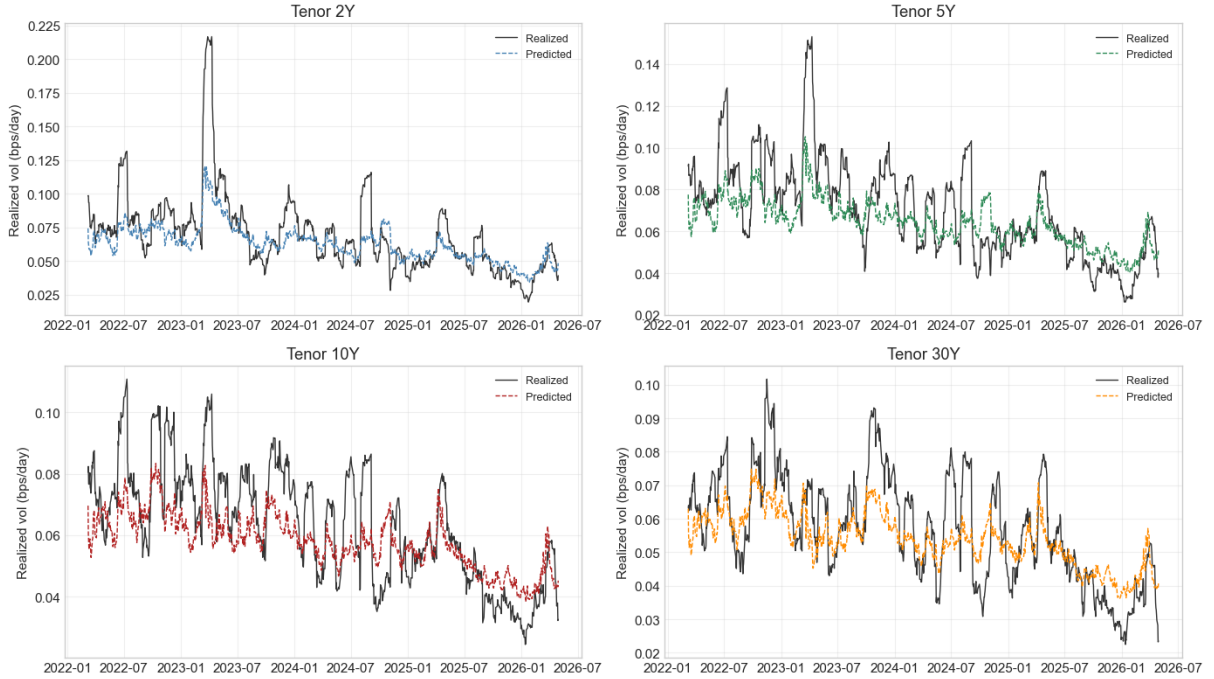


Figure 5: Elastic Net predicted vs. realized rate volatility accross tenors

6 Interpretation and Limitations

The central finding is that a regularised linear model with 41 economically motivated features produces reliable directional forecasts of the volatility term structure. $IC = 0.649$ on PC1 and directional accuracy exceeding 70% on two components are practically significant results. The failure of Gradient Boosting to add value is itself informative: the signal in the feature set is predominantly linear, and that the structure do not involve the kind of non-linear interactions that tree-based models are designed to exploit.

There are various practical applications for interest rate derivatives. A rising PC1 forecast signals a volatility shift across all tenors, creating an opportunity to accumulate long vega positions before the move. A positive PC2 signal indicates that short-tenor volatility (2Y) will outperform long-tenor (30Y) over the next month, justifying an accumulation of short-term vega buckets. These signals are most reliable in calm-to-moderate regimes: during stress events, the model's directional accuracy degrades.

Two limitations should be noted. First, realized volatility is used as a proxy for implied volatility — the variance risk premium implies a persistent bias. Second, stress-period underestimation is inherent to conditional mean regression. An extension to regime-switching specifications (HMM) or EGARCH-family models would directly address the fat-tail problem identified in the residual analysis.

References

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